iii. Less security measures.
iv. No flexibility.
v. There will be more noise in the working area.

iii. Multi motor drive

Many electric drives combined with a single common controlled structure is known as multi motor drive system.

a. Advantages

i. Each drive is powered by a single electric motor. It can only stop or use the required motor.
ii. Individual drives can be controlled.
iii. If any drive fails, it will not affect the other drives.

b. Applications

i. Rolling mills
ii. Paper mills
iii. Escalators
iv. Mixer metal works.

5.3 Selection of motors

Based on its speed, torque and accuracy, an electric machine should be chosen. It should be similar for the speed-torque characteristics of the machine loads.

Types of mechanical loads can be classified as follows:

5.3.1. Fixed speed loads

As shown in figure 5.9, some loads require steady speed when load varies. These are called fixed speed loads.

i. Mechanical Tools
ii. Hydraulic pumps
iii. Paper industries

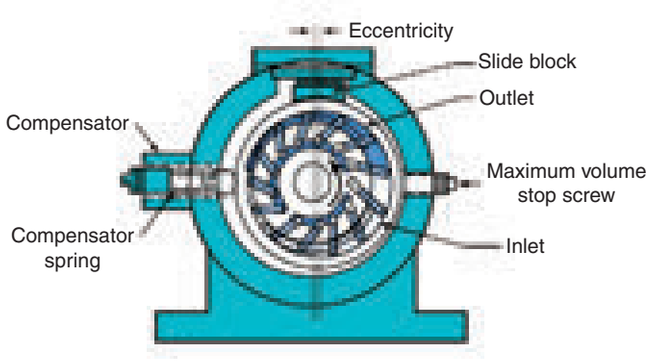


Fig. 5.9

5.3.2 Different speed loads

Some of the specific mechanical devices that has the same advantage are low speed, and torque requirement. These are in figure 5.10. Some drives require more speed limits.



Fig. 5.10

I. Crane
II. Hoist

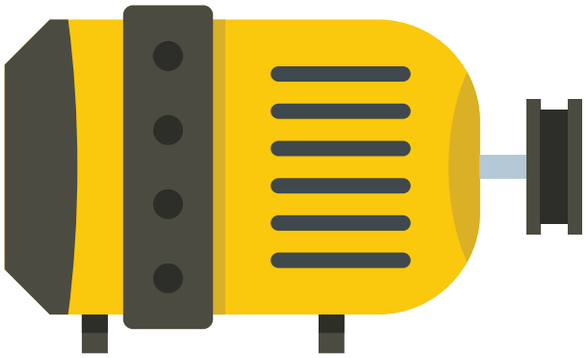


Fig. 5.11

BEE_ENG_Chapter_05.indd 79 12/27/2019 10:29:10 AM